

Ni-catalysed acceptorless dehydrogenative aromatisation of cyclohexanones enabled by concerted catalysis specific to supported nanoparticles

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The dehydrogenative aromatisation of cyclohexanone derivatives has had a transformative influence on the synthesis of aromatic compounds because functional groups can be easily introduced at desired positions via classic organic reactions without being limited by *ortho*-, *meta*- or *para*-orientations. However, research is still limited on acceptorless dehydrogenative aromatisation, especially with regard to nonprecious-metal catalysts. Ni is a promising candidate catalyst as a congener of Pd, but thermally Ni-catalysed dehydrogenative aromatisation has not been reported even in an oxidative manner because of the difficulty of β -hydride elimination and the fast re-insertion of Ni–H species. Here, we report a CeO₂-supported Ni(0) nanoparticle catalyst for acceptorless dehydrogenative aromatisation of cyclohexanone derivatives. This catalyst is widely applicable to various compounds such as cyclohexanols, cyclohexylamines, *N*-heterocycles, enamines and β -heteroatom-substituted ketones. Through various experiments, we demonstrate that the present reaction was achieved by the concerted catalysis utilizing metal ensembles unique to supported metal nanoparticle catalysts.

Metal ensembles of nanoparticles exhibit a catalytic performance distinct from that of mononuclear complexes because the substrates and intermediates can be activated or adsorbed on multiple active sites of the metal nanoparticles in a unique fashion^{1–4}. For example, in 2021, Ordonsky et al.⁴ reported a novel strategy for size-dependent selective hydrogenation using molecularly imprinted Pd nanoparticle catalysts, which they prepared via poisoning of the Pd surface except for positions adsorbed by template molecules such as benzene. This selectively catalysed hydrogenation of aromatic compounds with the same size as the template molecules. In addition, combining metal nanoparticles with appropriate supports realises a synergistic catalysis that

efficiently promotes the desired reaction while enhancing the stability of the active species^{5–8}.

Aromatic compounds are a ubiquitous building block in various fields, and many efforts have been made to establish methodologies for their synthesis. In general, substituted aromatic compounds are synthesised via classic nucleophilic aromatic substitution (S_NAr) or electrophilic aromatic substitution (S_EAr), but the regioselectivity is constrained by the substituents^{9,10}. Although cross-coupling and C–H functionalisation have been used to realise regioselective synthesis, inherent problems remain such as regioselectivity control of pre-functionalisation, generation of toxic (pseudo-)halogenated products and the requirement for directing groups^{11–13}. As an alternative

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approach, in 2011, Stahl et al.¹⁴ reported synthesising phenols from nonaromatic cyclohexanones via Pd-catalysed aerobic dehydrogenative aromatisation, where the substituents are easily introduced regioselectively in any position by classic organic reactions¹⁵. Since then, catalytic dehydrogenative aromatisation has been applied to synthesising various kinds of substituted aromatic hydrocarbons including phenols, anilines and ethers (Fig. 1a)^{16–19}.

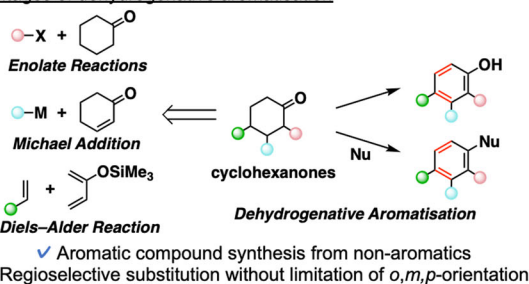
Dehydrogenative aromatisation is classified into two types: oxidative or acceptorless^{16–19}. Most reports on dehydrogenative aromatisation have been on the oxidative type, which mainly utilised Cu- or Pd-based catalysts or transition-metal-free systems. Research on acceptorless dehydrogenative aromatisation is scarce despite the high atom economy, where H₂ is the sole theoretical by-product in the case of phenol synthesis (Fig. 1a). Liu et al.²⁰ reported the first example of liquid-phase acceptorless dehydrogenative aromatisation of cyclohexanones to phenols using Pd/C. Since then, other groups, including ours, have also reported synthesising aromatic compounds by combining acceptorless dehydrogenative aromatisation with condensation between nucleophiles or electrophiles and cyclohexanones in the presence of Pd-based catalysts^{21–28}. Although noble-metal catalysts like Pd have been widely applied to acceptorless dehydrogenation^{29–31} as exemplified by such aromatisation reactions, their high cost and limited supply have increased demand for alternative approaches that use nonprecious-metal catalysts. In recent years, acceptorless dehydrogenation through hydrogen atom transfer (HAT), which utilises both photoredox catalysts and transition-metal catalysts including non-precious ones, has attracted much attention because of its high site selectivity and mild reaction conditions (Fig. 1b)^{15,32–39}. However, the reported photocatalytic systems for acceptorless dehydrogenative aromatisation typically require activated cyclohexane skeletons possessing benzylic, allylic or heteroatom-neighbouring C–H bonds, which have relatively low bond dissociation energies (BDEs) (Fig. 1b)^{15,34–37}. Quite recently, Kanai et al.³⁸ reported acceptorless dehydrogenative

aromatisation of cycloalkanes without requiring the aforementioned weak C–H bonds based on dual HAT strategy by utilizing photoredox catalysis, but this report did not apply to dehydrogenative aromatisation of cyclohexanones to afford phenols. Moreover, during the revision of this paper, Jin et al.³⁹ reported a decatungstate/cobalt dual photoredox catalytic system for acceptorless α,β -dehydrogenation of ketones applicable to dehydrogenative aromatisation of cyclohexanones, of which catalytic performance for the dehydrogenative aromatisation of cyclohexanones without possessing easily cleavable C–H bonds was not sufficient (substantial amounts of cyclohexenone intermediates remained). Thus, nonprecious-metal catalysts for the acceptorless dehydrogenative aromatisation of cyclohexanones are not established even using photoredox catalysts, and to our knowledge, thermally catalysed acceptorless dehydrogenative aromatisation of cyclohexanones by nonprecious-metal catalysts have not been reported albeit its practical utility.

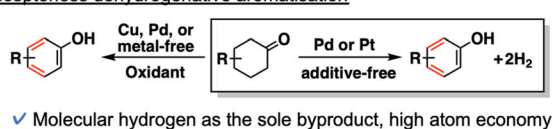
Previous reports on acceptorless dehydrogenative aromatisation have mainly used Pd-based catalysts. Ni is a congener of Pd, and thus, we considered it as a promising nonprecious-metal candidate for catalysing acceptorless dehydrogenative aromatisation. However, no Ni-based catalysts have been reported for dehydrogenative aromatisation without photoirradiation, even for the oxidative type. Using Ni-based catalysts to achieve acceptorless dehydrogenative aromatisation has several challenges compared with well-known Pd-based catalysts. First, the activation energy of β -hydride elimination to afford alkenes is higher in Ni-alkyl intermediates than in Pd-based counterparts because the lower electronegativity of Ni weakens the agostic interactions between Ni and H species at the β -position and because its smaller atomic radius renders the transition state more unstable due to geometrical limitations (Fig. 1c)^{40–43}. Second, the fast migratory insertion of hydride species on the Ni centre into alkene species^{44–46} before H₂ evolution inhibits the formation of dehydrogenated products (Fig. 1c). Finally, the Ni-based catalyst needs to be sufficiently stable to

a Dehydrogenative aromatisation of cyclohexanones

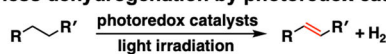
Advantages of dehydrogenative aromatisation



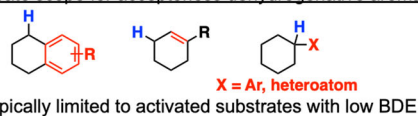
Acceptorless dehydrogenative aromatisation



b Acceptorless dehydrogenation by photoredox catalysts

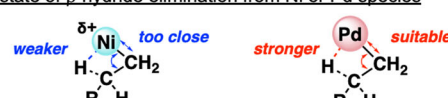


Limited substrate scope for acceptorless dehydrogenative aromatisation

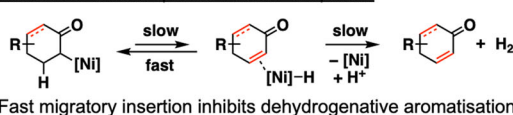


c Challenges for Ni-catalysed dehydrogenative aromatisation

Transition state of β -hydride elimination from Ni or Pd species



Migratory insertion of alkene species from Ni–H species



d This work

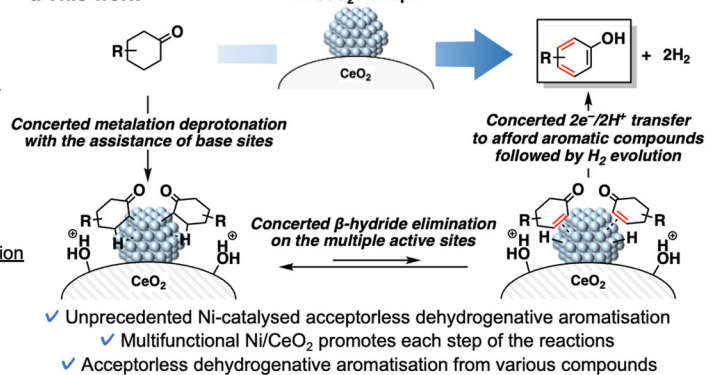


Fig. 1 | Background and overview of the study. **a** Advantages and types of dehydrogenative aromatisation of cyclohexanones. **b** Acceptorless dehydrogenation by photoredox catalysts. **c** Challenges for Ni-catalysed dehydrogenative aromatisation.

d This work: First example of Ni-catalysed acceptorless dehydrogenative aromatisation of various compounds including cyclohexanones facilitated by concerted catalysis utilizing the metal ensembles and support basicity of Ni/CeO₂-NaNaph.

function even at high temperatures because acceptorless dehydrogenative aromatisation is more thermodynamically unfavourable than oxidative aromatisation.

Herein, we have developed acceptorless dehydrogenative aromatisation of cyclohexanones to phenols by utilizing a multifunctional CeO₂-supported Ni(O) nanoparticle heterogeneous catalyst prepared via reduction using sodium naphthalenide (Ni/CeO₂-NaNaph). Unprecedented Ni-catalysed dehydrogenative aromatisation was enabled by the following features (Fig. 1d): (i) The concerted metalation deprotonation of cyclohexanones to Ni(O) nanoparticles with the assistance of Brønsted base sites in the CeO₂ support without the addition of external bases, (ii) the concerted β -hydride elimination to form cyclohexenones and Ni–H species on multiple active sites of Ni nanoparticles to overcome the aforementioned disadvantages derived from its smaller atomic radius for β -hydride elimination and (iii) the concerted two-electron/two-proton transfer from one cyclohexenone to another cyclohexenone (fast disproportionation of cyclohexenones) to form phenols and cyclohexanones enabled by their simultaneous adsorption on the surface of Ni nanoparticles, which inhibits the re-insertion of the Ni–H species to cyclohexenones, followed by H₂ evolution from the Ni–H species on Ni nanoparticles and protons on CeO₂. This system was also applicable to acceptorless dehydrogenation of various compounds such as cyclohexanols, cyclohexylamines, *N*-heterocycles, enamines and β -heteroatom-substituted ketones. We thoroughly performed catalyst characterisation, kinetic analyses and control experiments to demonstrate the unique performance of the prepared catalytic system and to compare it to other catalysts.

Results

Catalyst characterisation

Ni/CeO₂-NaNaph was prepared as described in our previous report with some modifications^{47,48}. Briefly, a CeO₂-supported Ni hydroxide precursor (Ni(OH)_x/CeO₂) prepared via deposition–precipitation was reduced by sodium naphthalenide (NaNaph) in tetrahydrofuran (THF) at room temperature (–24 °C) in an Ar atmosphere to afford Ni/CeO₂-NaNaph (Ni: 1.38 wt%). The Ni K-edge X-ray absorption near-edge structure (XANES) spectrum of Ni/CeO₂-NaNaph was between those of Ni foil and Ni(OH)₂, which indicates that most Ni species in Ni/CeO₂-NaNaph were zero-valent; according to linear combination fitting (LCF), ~76% of the Ni species were zero-valent (Fig. 2a; Supplementary Fig. 1a). This is mostly consistent with the X-ray photoelectron spectroscopy (XPS) results (Fig. 2b; Supplementary Fig. 2, Supplementary Table 1). The *k*³-weighted Fourier-transformed Ni K-edge-extended X-ray absorption fine-structure (EXAFS) spectrum of Ni/CeO₂-NaNaph exhibited scattering originated from Ni–Ni bond of Ni metal in the first coordination sphere while very weak counterparts in the second coordination sphere were observed, suggesting the presence of highly dispersed Ni(O) species in Ni/CeO₂-NaNaph (Fig. 2c; Supplementary Fig. 3, Supplementary Table 2). In fact, the diffuse reflectance infrared Fourier transform (DRIFT) spectrum of CO adsorbed on Ni/CeO₂-NaNaph exhibited a peak around 2050 cm^{–1}, which could be assigned to linear CO species adsorbed on zero-valent Ni species, but bridged species were not observed (Fig. 2d)⁴⁹. Although direct observation of Ni species supported on CeO₂ by high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) was difficult (Fig. 2e), STEM-energy-dispersive spectroscopy (EDS) mapping confirmed that highly dispersed Ni species were supported on CeO₂ (Fig. 2f, g). The X-ray diffraction (XRD) patterns of Ni/CeO₂-NaNaph were comparable with those of CeO₂ and Ni(OH)_x/CeO₂, which supports that Ni(O) species were successfully immobilised on CeO₂ with the CeO₂ structures intact (Fig. 2h).

Catalyst effect

We initially investigated the effect of supported nonprecious-metal nanoparticle catalysts on acceptorless dehydrogenative aromatisation

using 4-methylcyclohexanone (**1a**) as a model substrate under the reaction conditions indicated in Table 1. With Ni/CeO₂-0.3NaNaph (prepared in the same manner as in our previous report⁴⁷ by using 0.3 equivalent of NaNaph to the amount of NaNaph used to prepare Ni/CeO₂-NaNaph), the desired dehydrogenated product *p*-cresol (**2a**) was obtained with a 50% yield along with 2-methylated phenol (**2a'**) with a 3% yield (Table 1, entry 1) (**2a'** formation mechanism was discussed in the Supplementary Information). Interestingly, other supported 3d-transition-metal nanoparticle catalysts including Cu, Co, Mn, Fe and Zn did not catalyse dehydrogenative aromatisation (Table 1, entries 2–6). In addition, as-purchased CeO₂, CeO₂ reduced by NaNaph (CeO₂-NaNaph) and Ni(OH)_x/CeO₂ did not catalyse the reaction (Table 1, entries 7–9). These results indicate that the reduced Ni was the actual active species. In fact, while the examined Ni loading did not greatly affect dehydrogenative aromatisation (Supplementary Table 3), the yield of **2a** was slightly improved by increasing the amount of reducing reagent to prepare Ni/CeO₂-NaNaph (Table 1, entries 10–12) probably due to the increased Ni(O) active species using higher concentration of NaNaph, which was revealed by the XPS spectra (Supplementary Figs. 2 and 4, Supplementary Table 1). Considering the DRIFT spectra of CO adsorbed on the catalysts (Supplementary Fig. 5), whose peak area indicate the amount of surface Ni(O) active sites, Ni/CeO₂-NaNaph was determined to be the optimized catalyst because Ni species could be sufficiently reduced without covering many active sites with reductant-derived substances like naphthalene when utilizing moderate NaNaph concentration. Next, we investigated the effect of the supports. Only Ni/CeO₂-NaNaph exhibited a high catalytic performance to afford **2a** with a 74% yield (Table 1, entry 11). Ni catalysts on other supports such as hydroxyapatite (HAP), ZrO₂, TiO₂ and Al₂O₃ hardly catalysed the reaction (Table 1, entries 13–16). When Ni(cod)₂ and IPr (1,3-bis(2,6-diisopropylphenyl)imidazol-2-ylidene) (Ni-IPr complex) were used as the catalyst, the dehydrogenation hardly occurred either (Table 1, entry 17). The solvent was also shown to have a substantial effect, and the yield of **2a** was highest when *N,N*-dimethylacetamide (DMA) was used as the solvent (Supplementary Table 4).

Dehydrogenative aromatisation of 3,5-dimethylcyclohexanone (**1g**) was immediately stopped by hot filtration of Ni/CeO₂-NaNaph (Supplementary Fig. 6), and inductively coupled plasma atomic emission spectroscopy (ICP-AES) showed that almost no Ni species were in the filtrate (Ni: 0.21% of the Ni species used for the reaction), which indicates that the observed catalysis was heterogeneous. Although Ni/CeO₂-NaNaph could be reused after catalyst regeneration via calcination at 300 °C in an air to remove the adsorbed phenol products followed by reduction using NaNaph, repeated use of the catalyst is difficult because of the aggregation of Ni(O) nanoparticles and decreased Ni(O) ratio during the regeneration (Supplementary Figs. 7–10). Then, we investigated several kinds of catalyst washing methods for removing the adsorbed phenols instead of the calcination in an air. As a result, the final **2a** and **2a'** yields obtained by the used catalyst treated with potassium *tert*-butoxide in methanol under an Ar atmosphere (Ni/CeO₂-NaNaph-KO^{*t*}Bu) were comparable to those by fresh Ni/CeO₂-NaNaph (Supplementary Fig. 11) due to the successful removal of adsorbed phenols (Supplementary Fig. 12) (see Supplementary Information for the detailed discussion on reuse tests).

Role of the CeO₂ support

To clarify the role of the CeO₂ support, Ni/CeO₂-NaNaph, Ni/HAP-NaNaph or Ni-IPr complex was used as the catalyst for the present dehydrogenation with additives such as acid (PhCOOH, benzoic acid), base (represented as Na₂CO₃) and CeO₂-NaNaph (Table 1, entries 18–21, Supplementary Table 5). First, the effect of the basicity of the supports on dehydrogenative aromatisation was investigated. Dehydrogenative aromatisation of **1a** catalysed by Ni/CeO₂-NaNaph or Ni/HAP-NaNaph was completely suppressed in the presence of PhCOOH (Table 1, entry 18, Supplementary Table 5, entry 8). Meanwhile,

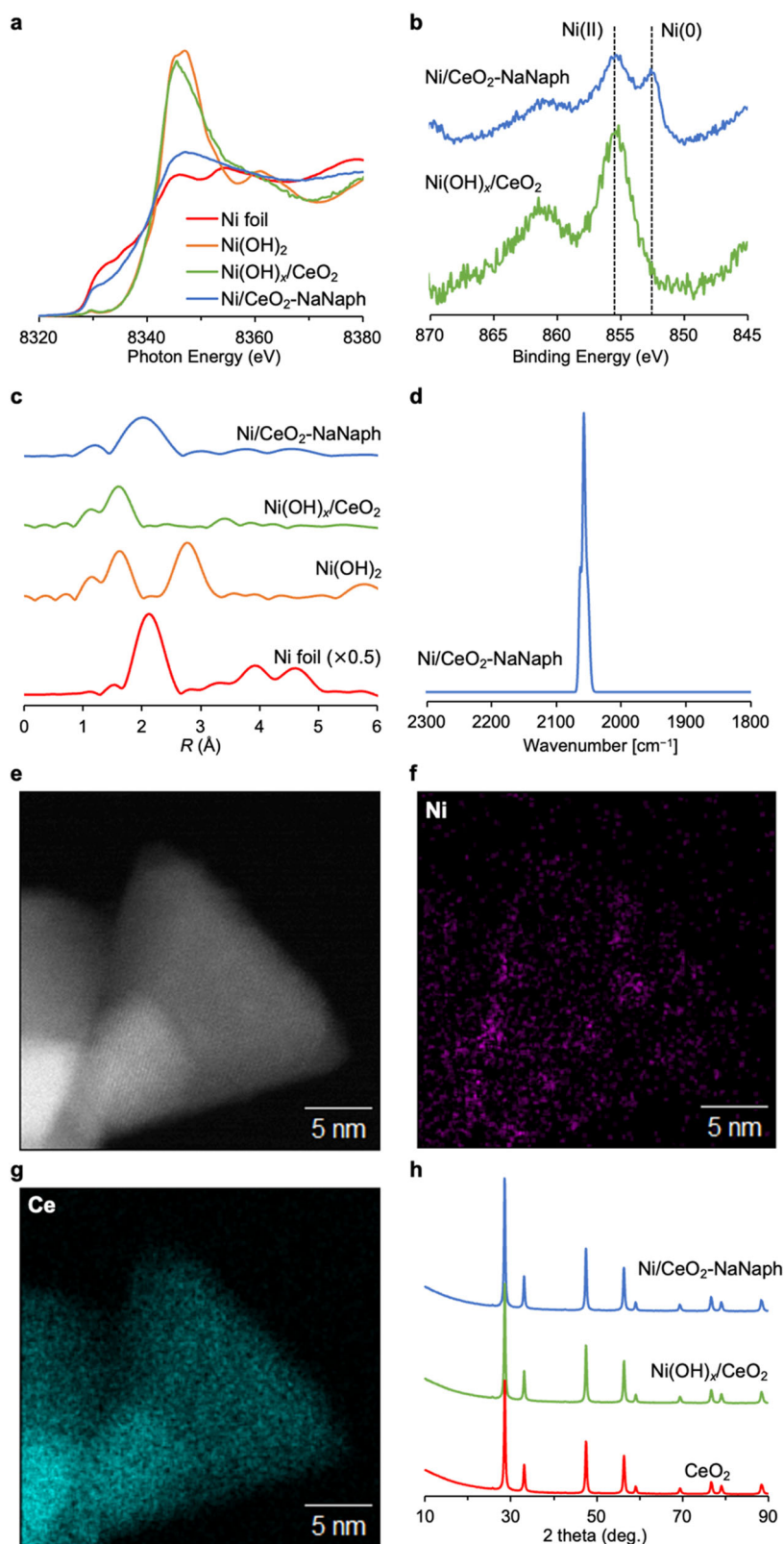


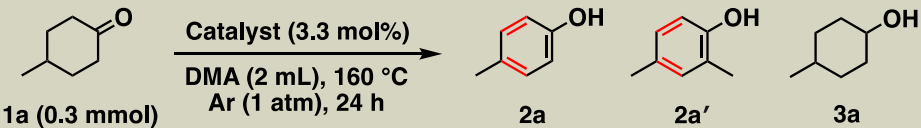
Fig. 2 | Characterisation of Ni catalysts. **a** Ni K-edge XANES spectra. **b** Ni 2p XPS spectra. **c** k^3 -weighted Fourier-transformed Ni K-edge EXAFS spectra. **d** DRIFT spectrum of CO adsorbed on Ni/CeO₂-NaNaph. **e** HAADF-STEM image of Ni/CeO₂-

NaNaph. STEM-EDS mappings of Ni/CeO₂-NaNaph showing the distributions of **(f)** Ni (magenta) and **(g)** Ce (cyan). **h** XRD patterns.

dehydrogenative aromatisation of **1a** using Ni/HAP-NaNaph was promoted by the addition of CeO₂-NaNaph and Na₂CO₃, which indicates that the basicity of CeO₂ promoted the deprotonative metalation step (Table 1, entries 19–21, Supplementary Table 5, entries 6, 7 and 9).

Then, we further investigated the effect of other bases on the acceptorless dehydrogenative aromatisation of **1a** in the presence of Ni/HAP-NaNaph (Supplementary Table 6), which revealed that weak bases like Li₂CO₃ were effective for promoting this reaction while strong bases

Table 1 | Effects of catalysts on the dehydrogenative aromatisation of 4-methylcyclohexanone (1a)^a

				
Entry	Catalyst	Yield (%)		
		2a	2a'	3a
1	Ni/CeO ₂ -0.3NaNaph	50	3	13
2	Cu/CeO ₂ -0.3NaNaph	1	<1	<1
3	Co/CeO ₂ -0.3NaNaph	<1	<1	<1
4	Mn/CeO ₂ -0.3NaNaph	<1	<1	<1
5	Fe/CeO ₂ -0.3NaNaph	<1	<1	<1
6	Zn/CeO ₂ -0.3NaNaph	<1	<1	<1
7	CeO ₂	<1	<1	<1
8	CeO ₂ -NaNaph	<1	<1	<1
9	Ni(OH) ₂ /CeO ₂	<1	<1	<1
10 ^b	Ni/CeO ₂ -0.3NaNaph	52	3	5
11 ^b	Ni/CeO ₂ -NaNaph	74	13	3
12 ^b	Ni/CeO ₂ -1.7NaNaph	72	9	3
13 ^b	Ni/HAP-NaNaph	1	<1	<1
14 ^b	Ni/ZrO ₂ -NaNaph	<1	<1	<1
15 ^b	Ni/TiO ₂ -NaNaph	<1	<1	<1
16 ^b	Ni/Al ₂ O ₃ -NaNaph	<1	<1	<1
17 ^c	Ni(cod) ₂ + IPr	3	<1	<1
18 ^d	Ni/CeO ₂ -NaNaph	<1	<1	<1
19	Ni/HAP-NaNaph	2	<1	<1
20 ^e	Ni/HAP-NaNaph	9	<1	<1
21 ^f	Ni/HAP-NaNaph	8	<1	<1

^a Reaction conditions: **1a** (0.3 mmol), catalyst (3.3 mol%), DMA (2 mL), Ar (1 atm), 160 °C, 24 h. Yields were determined by GC analysis using *n*-hexadecane as an internal standard.^b 180 °C.^c IPr (6.7 mol%).^d PhCOOH (0.3 mmol).^e CeO₂-NaNaph (40 mg).^f Na₂CO₃ (0.3 mmol).

like Cs₂CO₃ and NaOH inhibited this reaction probably because of side reactions such as aldol condensation of **1a** and decomposition of the solvent (*N,N*-dimethylacetamide, DMA) accelerated by strong bases (Supplementary Table 6, entries 1–11). Organic bases such as 1,8-diazabicyclo[5.4.0]-7-undecene (DBU), 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD) and 4-dimethylaminopyridine (DMAP) also inhibited this reaction (Supplementary Table 6, entries 12–14). Considering that aldol condensation byproducts and DMA-derived byproducts were not observed differently from the case of inorganic bases, the preferential adsorption of organic bases on the catalyst over **1a** is the plausible reason for the inhibition.

In the case of Ni/CeO₂-NaNaph, similarly, the addition of Na₂CO₃ or Li₂CO₃ accelerated the initial rate of **1a** acceptorless dehydrogenative aromatisation (Supplementary Fig. 13), but the **2a** yields after the reactions for 24 h slightly decreased, especially with Na₂CO₃ (Supplementary Table 5, entries 1–3). The treatment of acetic acid (CH₃COOH) after the reaction using Ni/CeO₂-NaNaph and Li₂CO₃ improved the **2a** yield a little whereas the decreased **2a** yield with Na₂CO₃ addition was not improved by the CH₃COOH treatment (Supplementary Fig. 14). Therefore, the neutralization reaction between **2a** and Li₂CO₃ is the possible reason for the slight decrease of the final **2a** yield by Li₂CO₃ and the aforementioned side reactions caused by the stronger base Na₂CO₃ likely decreased the final **2a** yield. On the other hand, the addition of CeO₂-NaNaph to Ni/CeO₂-NaNaph

decreased both the initial production rate and the final yield of **2a** (Supplementary Fig. 13, Supplementary Table 5) possibly because of the inhibition effect of residual naphthalene in CeO₂-NaNaph and because of adsorption of **2a** on CeO₂-NaNaph confirmed by infrared spectra using the attenuated total reflection method (ATR-IR) of the physical mixture catalyst retrieved after the reaction (Supplementary Fig. 15). From the above, Ni/CeO₂-NaNaph possessed suitable weak basicity like Li₂CO₃ for the promotion of the desired dehydrogenation without side reactions caused by stronger bases and the addition of Li₂CO₃ can increase the basic sites necessary for the dehydrogenation to enhance the production rate of **2a** with both the Ni catalyst and produced **2a** almost intact.

Next, when Ni-IPr complex was used as the catalyst in the presence of the additives, PhCOOH and Na₂CO₃ inhibited dehydrogenative aromatisation of **1a**, but the **2a** yield was slightly improved in the presence of CeO₂-NaNaph (Supplementary Table 5, entries 10–13), suggesting that CeO₂ could play another role in addition to the aforementioned base in the present dehydrogenation system. The DRIFT spectrum of CO adsorbed on the retrieved CeO₂-NaNaph after the reaction with Ni-IPr complex revealed that slight amounts of Ni(0) species were deposited on CeO₂ during the reaction (Supplementary Fig. 16). Thus, the Ni species immobilized on CeO₂ likely functioned as the catalyst to improve the **2a** yield. On the other hand, the addition of HAP instead of CeO₂-NaNaph to the Ni-IPr complex-catalysed **1a**

aromatisation did not improve the reaction results (Supplementary Table 5, entry 14), indicating stronger interaction between Ni(O) nanospecies and CeO₂⁴⁷. In fact, the *k*³-weighted Ni K-edge EXAFS spectrum of Ni/CeO₂-NaNaph exhibited very weak scattering originating from the Ni–Ni bond in the second coordination sphere whereas that of Ni/HAP-NaNaph exhibited relatively strong scattering (Supplementary Fig. 17, Supplementary Table 2). These results indicate that the strong interaction between Ni(O) nanospecies and CeO₂ induced the formation of ultrasmall Ni(O) nanospecies in the Ni/CeO₂-NaNaph. Thus, the exceptional catalytic activity of Ni/CeO₂-NaNaph for dehydrogenative aromatisation can be attributed to the synergistic catalysis between Ni(O) species and the basic sites of CeO₂ and the high dispersion of Ni(O) species induced by strong interaction with CeO₂.

Substrate scope

Next, we evaluated the substrate scope of Ni/CeO₂-NaNaph for catalysing acceptorless dehydrogenative aromatisation of cyclohexanones (**1**) to phenols (**2**) (Fig. 3, **2a–2v** from cyclohexanones; see Supplementary Table 7 for the detailed information on the by-products in substrate scope investigation). This catalytic system is applicable to *para*-, *meta*- and *ortho*-methyl-substituted phenol synthesis (**1a–1c**). Other alkyl-substituted cyclohexanones including 4-ethyl- (**1d**), 4-propyl- (**1e**), 3,4-dimethyl- (**1f**), 3,5-dimethyl- (**1g**), 4-*tert*-butyl- (**1h**) and 4-cyclohexyl- (**1i**) ones or non-substituted cyclohexanone (**1j**) were successfully converted into the corresponding phenols with good to moderate yields. This system could also be applied to the dehydrogenative aromatisation of aryl-substituted cyclohexanones (**1k, 1l**). In the case of cyclohexanones with various functional groups such as phenolic hydroxy, acetal, ethyl ester and acetamide, this catalytic system worked well at affording the desired dehydrogenated products (**1m–1q**) while a 2-allyl group was hydrogenated during the reaction (**1r**). In addition, 1-tetralone (**1s**) and 1,4-cyclohexanedione (**1t**) were converted into corresponding phenols with good yields. Cyclohexenones such as 3-methyl-2-cyclohexen-1-one (**1u**) and carvone (**1v**) could be subjected to dehydrogenative aromatisation, but the C=C bond in the corresponding phenol obtained from carvone was hydrogenated. On the other hand, several functional groups, especially strong electron-withdrawing groups, could not be applied to the present acceptorless dehydrogenative aromatisation (Supplementary Fig. 18), possibly due to the difficulty in the β-hydride elimination (the turnover-limiting step as confirmed later) from electron-deficient substrates.

Several studies have reported on the acceptorless dehydrogenation of secondary alcohols to afford the corresponding ketones in the presence of supported Ni nanoparticle catalysts^{50,51}, which motivated us to investigate acceptorless dehydrogenative aromatisation of cyclohexanols (**3**) to phenols (**2**). Cyclohexanol is commercially obtained as ketone–alcohol (K–A) oil via hydrogenation of benzene followed by aerobic cyclohexane oxidation. Thus, if H₂ formed by acceptorless dehydrogenative aromatisation of cyclohexanol is retrieved and used for benzene hydrogenation, then a formal process of aerobic benzene oxidation to phenol could be constructed²¹. As expected, 4-methylcyclohexanol (**3a**) was successfully converted into the corresponding phenol (**2a**) with a good yield in the presence of Ni/CeO₂-NaNaph via dehydrogenation of **3a** to **1a** followed by dehydrogenative aromatisation (Supplementary Fig. 19) while a CeO₂-supported Pd nanoparticle catalyst (Pd/CeO₂-NaNaph) produced **2a** with a much lower yield (Supplementary Table 8). This is because Pd nanoparticles hardly catalyse the dehydrogenation of cyclohexanols to cyclohexanones. In fact, acceptorless dehydrogenation of 3-decanol to 3-decanone hardly proceeded in the presence of Pd/CeO₂-NaNaph whereas Ni/CeO₂-NaNaph efficiently catalysed the acceptorless alcohol dehydrogenation (Supplementary Fig. 20). The substrate scope of the catalytic system for dehydrogenative aromatisation starting from cyclohexanols (Fig. 3, **2a–2d**, **2f–2h**, **2k**, **2l**, **2p**, **2s**, **2t**, **2w** from

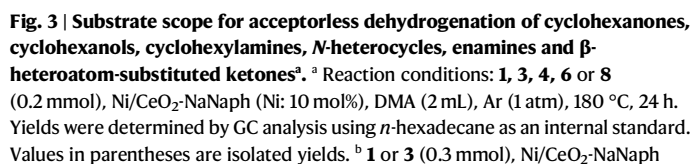
cyclohexanols) includes various alkyl-, phenyl-, acetamide- or dimethylamino-substituted cyclohexanols (**3a–3d**, **3f–3h**, **3k**, **3l**, **3p**, **3w**). In addition, 1,2,3,4-tetrahydro-1-naphthol (**3s**) and 1,4-cyclohexanediol (**3t**) were efficiently converted into phenols.

We found that Ni/CeO₂-NaNaph could also catalyse the acceptorless dehydrogenative aromatisation of cyclohexylamines (**4a–4d**) to afford primary anilines (Fig. 3, **5a–5d**), enamines (**6a–6c**) to afford aryl amines (Fig. 3, **7a–7d**) and *N*-heterocycles (**8a–8d**) (Fig. 3, **9a–9d**). With regard to dehydrogenative aromatisation for enamines, using Ni/CeO₂-NaNaph to catalyse dehydrogenative aromatisation from 1-morpholino-cyclohexene (**6a**) resulted in the gradual formation of a dehydrogenated product (**7a**) via sequential dehydrogenation (Supplementary Fig. 21a) to achieve a high yield of **7a** (Fig. 3), whereas using Pd/CeO₂-NaNaph as the catalyst produced **7a** and the hydrogenated product (**7a'**) at a ratio of 1:2 immediately after the reaction was started due to the disproportionation of **6a** on Pd nanoparticles⁵² (Supplementary Fig. 21b), which resulted in a moderate yield of **7a** even after 24 h (Supplementary Fig. 22). Then, Ni/CeO₂-NaNaph-catalysed α,β-dehydrogenation of *N*-alkyl piperidones (**10a–10c**) was conducted (Fig. 3, **11a–11c**), which surprisingly selectively converted 1-methyl-4-piperidone (**10a**) into 1-methyl-4(*1H*)-pyridinone (**11a**) with an excellent yield via double dehydrogenation of **10a**. This reaction has not previously been reported. The reactions from 1-methyl-4-piperidone with a methyl group at the α-position of the carbonyl group (**10b**) and 1-ethyl-4-piperidone (**10c**) were successful, and they afforded corresponding double-dehydrogenated products with good yields. Moreover, flavanone (**10d**), a kind of β-heteroatom-substituted ketones, was dehydrogenated to flavone (Fig. 3, **11d**) in a moderate yield. In addition, this catalytic system was also applicable to one-pot reactions composed of condensation and Ni/CeO₂-NaNaph-catalysed acceptorless dehydrogenative aromatisation, affording an *o*-alkylated phenol or an *N,N*-disubstituted aniline (Fig. 4).

Mechanistic studies

To confirm that the dehydrogenative aromatisation proceeded without any hydrogen acceptors, gas chromatography (GC) was used to analyse the gas phase after dehydrogenative aromatisation of **1g** (Supplementary Fig. 23). The results showed approximately two equivalents of H₂ to the dehydrogenated product (**2g**), which indicates that the dehydrogenative aromatisation was the acceptorless type. In addition, when 3-methyl-2-cyclohexen-1-one (**1u**) was the substrate, *m*-cresol (**2b**) and 3-methylcyclohexanone (**1b**) were produced with almost the same yields 10 min after the reaction was started (Supplementary Fig. 24). Together with the above results on support effects and our previous reports on acceptorless dehydrogenative aromatisation by supported Pd nanoparticle catalysts^{21–27,52,53}, we propose the following mechanism for the formation of phenols from cyclohexanones (Supplementary Fig. 25): (1) cyclohexanones adsorb onto the surface of Ni nanoparticles; (2) cyclohexanones undergo concerted metalation deprotonation to Ni species with the assistance of Brønsted base sites in CeO₂; (3) β-hydride elimination takes place to afford cyclohexenones; (4) cyclohexenones undergo disproportionation to afford phenols and cyclohexanones on the surface of the Ni nanoparticles and (5) H₂ evolution occurs to achieve the catalytic cycle.

A series of kinetic analyses were conducted to determine the turnover-limiting step. We confirmed that the initial rate for the dehydrogenative aromatisation of cyclohexanone (**1j**) was saturated to the initial substrate concentrations under normal conditions in the present catalytic system (Supplementary Fig. 26), which indicates that the adsorption of **1j** to Ni species was not the turnover-limiting step. The **2b** production rate using **1u** as the starting material was much higher than that using **1b** (Supplementary Fig. 27), which indicates that the disproportionation of cyclohexenones was not the turnover-limiting step. The kinetic isotope effect (KIE) was not observed when **1j**



and cyclohexanone- α -d₄ (**1j**- α -d₄, k_H/k_{D4} =1.1) were used, which indicates that deprotonative metalation and H₂ evolution were not the turnover-limiting steps (Fig. 5a). In contrast, KIE was observed when **1j**- α -d₄ and cyclohexanone-d₁₀ (**1j**-d₁₀, k_{D4}/k_{D10} =2.3) were used (Fig. 5a). These results indicate that the turnover-limiting step of Ni/CeO₂-

NaNaph-catalysed dehydrogenative aromatisation is β -hydride elimination (as mentioned above, the addition of Na_2CO_3 accelerated the initial rate of **1a** acceptorless dehydrogenative aromatisation using Ni/CeO₂-NaNaph (Supplementary Fig. 13) possibly because the concentration of coordinating intermediates before the turnover-limiting

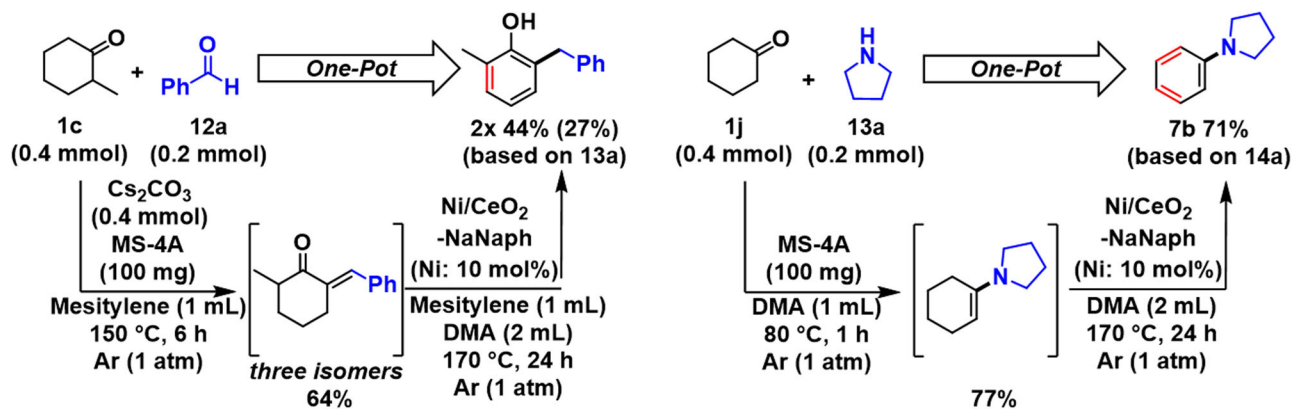


Fig. 4 | One-pot reactions composed of condensation and Ni/CeO₂-NaNaph-catalysed acceptorless dehydrogenative aromatisation. Reaction conditions were indicated in this figure. The second step reactions were carried out using the

filtrate after the first step reactions. Yields were determined by GC analysis using *n*-hexadecane as an internal standard. Values in parentheses are isolated yields. MS-4A molecular sieves 4 Å.

β -hydride elimination step increased by promoting deprotonative coordination of **1a**). On the other hand, in the presence of Pd/CeO₂-NaNaph, KIE was observed with **1j** and **1j- α -d₄** (k_H/k_{D4} = 2.6) but not with **1j- α -d₄** and **1j-d₁₀** (k_{D4}/k_{D10} = 1.3) (Fig. 5b), which indicates that the turnover-limiting step of Pd/CeO₂-NaNaph-catalysed dehydrogenative aromatisation is the deprotonative metalation of cyclohexanones.

We conducted various control experiments to elucidate why our catalytic system enabled unprecedented Ni-catalysed acceptorless dehydrogenative aromatisation. First, we compared Ni/CeO₂-NaNaph with other Ni complex catalysts for the dehydrogenative aromatisation of **1a** (Supplementary Table 9). While **2a** was not produced in the presence of Ni(II) complexes (Supplementary Table 9, entries 1–4), **2a** was slightly produced when Ni(cod)₂ and IPr were used (Table 1, entry 17). To clarify which steps were difficult for the Ni complexes, several experiments using deuterium-labelled cyclohexanones were conducted. The H/D exchange of **1j- α -d₄** efficiently proceeded 12 h after dehydrogenative aromatisation was started in the presence of the Ni-IPr complex (Fig. 5c), whereas the use of either Ni(cod)₂ or IPr caused the H/D exchange of **1j- α -d₄** to hardly proceed under the same reaction conditions (Supplementary Fig. 28). Thus, α -C–H activation of **1j** could occur with the Ni-IPr complex. However, the GC-MS and ²H NMR spectra of **1j- β , γ -d₆** (deuteration rate of α -position = 4%) after stirring with the Ni-IPr complex at 160 °C for 12 h were comparable to those before stirring, which indicates that a decrease in D from **1j- β , γ -d₆** or H/D scrambling between the α - and β -positions via β -hydride elimination followed by insertion did not occur (Fig. 5d; Supplementary Figs. 29–31). These results indicate that β -hydride elimination was difficult to induce in the case of Ni complexes. Based on these results and the different turnover-limiting steps of Ni nanoparticles and Pd nanoparticles, we concluded that Ni species generally have difficulty with β -hydride elimination because of their smaller atomic radius compared with Pd species^{40–43}, but the concerted catalysis by multiple active sites (Ni–Ni metal ensembles) of Ni nanoparticles could help attenuate the steric restriction and facilitate β -hydride elimination, which explains how our catalytic system achieved the unprecedented Ni-catalysed dehydrogenative aromatisation (Supplementary Fig. 32).

As noted earlier, when **1u** was used as the starting material, fast disproportionation occurred to afford **2b** and **1b** in the presence of Ni/CeO₂-NaNaph. However, the Ni-IPr complex or Ni(OH)₂/CeO₂ hardly catalysed the reaction (Fig. 5e, Supplementary Fig. 33). Considering the different mechanisms of Pd mononuclear active sites and Pd nanoparticles for dehydrogenative aromatisation to phenols^{53,54}, such a fast disproportionation of cyclohexanones to phenols and cyclohexanones appears specific to metal nanoparticle catalysts. This may be because the disproportionation proceeds via a concerted two-electron/two-proton transfer from one cyclohexenone to another cyclohexenone

after the two cyclohexenone molecules adsorb together on metal nanoparticles with the metal ensembles (Supplementary Fig. 34). Furthermore, acceptorless dehydrogenation of 4,4-dimethylcyclohexanone (**1y**), which affords nonaromatic dehydrogenated products even after double dehydrogenation, did not proceed with Ni/CeO₂-NaNaph, but small amounts of the corresponding dehydrogenated products 4,4-dimethylcyclohexa-2,5-dien-1-one (**2y**) and 4,4-dimethylcyclohex-2-en-1-one (**2y'**) formed in the presence of Pd/CeO₂-NaNaph (Fig. 5f) (Na₂CO₃ was required for **1y** dehydrogenation to promote the deprotonative coordination step and the H₂ evolution step because the re-insertion of Pd–H species to the non-aromatic dehydrogenated products occurred even in the case of Pd/CeO₂-NaNaph (Supplementary Table 10)). Based on the facileness of migratory insertion of Ni–H species to olefins compared with that of Pd–H species^{44–46}, these results indicate that the re-insertion of Ni–H species produced by β -hydride elimination of **1y** to **2y'** occurs faster than H₂ evolution, which explains the lack of dehydrogenated products from **1y** with Ni/CeO₂-NaNaph. Therefore, the fast disproportionation of cyclohexanones via the concerted two-electron/two-proton transfer on Ni nanoparticles to produce phenols, which are difficult to insert into Ni–H species, is another key factor for how we achieved Ni-catalysed acceptorless dehydrogenative aromatisation.

Based on the aforementioned results, the detailed proposed catalytic cycle of the present acceptorless dehydrogenative aromatisation of cyclohexanones is as follows (Fig. 5g); (1) coordination of cyclohexanones to the supported Ni nanoparticles, (2) concerted metalation deprotonation of cyclohexanones to Ni nanoparticles with the assistance of base sites on CeO₂, (3) concerted β -hydride elimination enabled by multiple active sites on Ni nanoparticles, (4) fast disproportionation to produce phenols and cyclohexanones via concerted two-electron/two-proton transfer from one cyclohexenone to another cyclohexenone on Ni nanoparticles and (5) H₂ evolution to achieve the catalytic cycle.

Discussion

We demonstrated the first example of Ni-catalysed acceptorless dehydrogenative aromatisation of cyclohexanones via α,β -dehydrogenation. Our CeO₂-supported Ni(0) nanoparticle catalyst is applicable to the acceptorless dehydrogenation of various substrates including cyclohexanols, cyclohexylamines, *N*-heterocycles, enamines and β -heteroatom-substituted ketones. The results of our catalyst characterisation, kinetic analyses and control experiments showed that Ni/CeO₂-NaNaph is a multifunctional catalyst that exhibits the concerted catalysis utilizing metal ensembles and support basicity to promote each step of acceptorless dehydrogenative aromatisation. Our results show how the

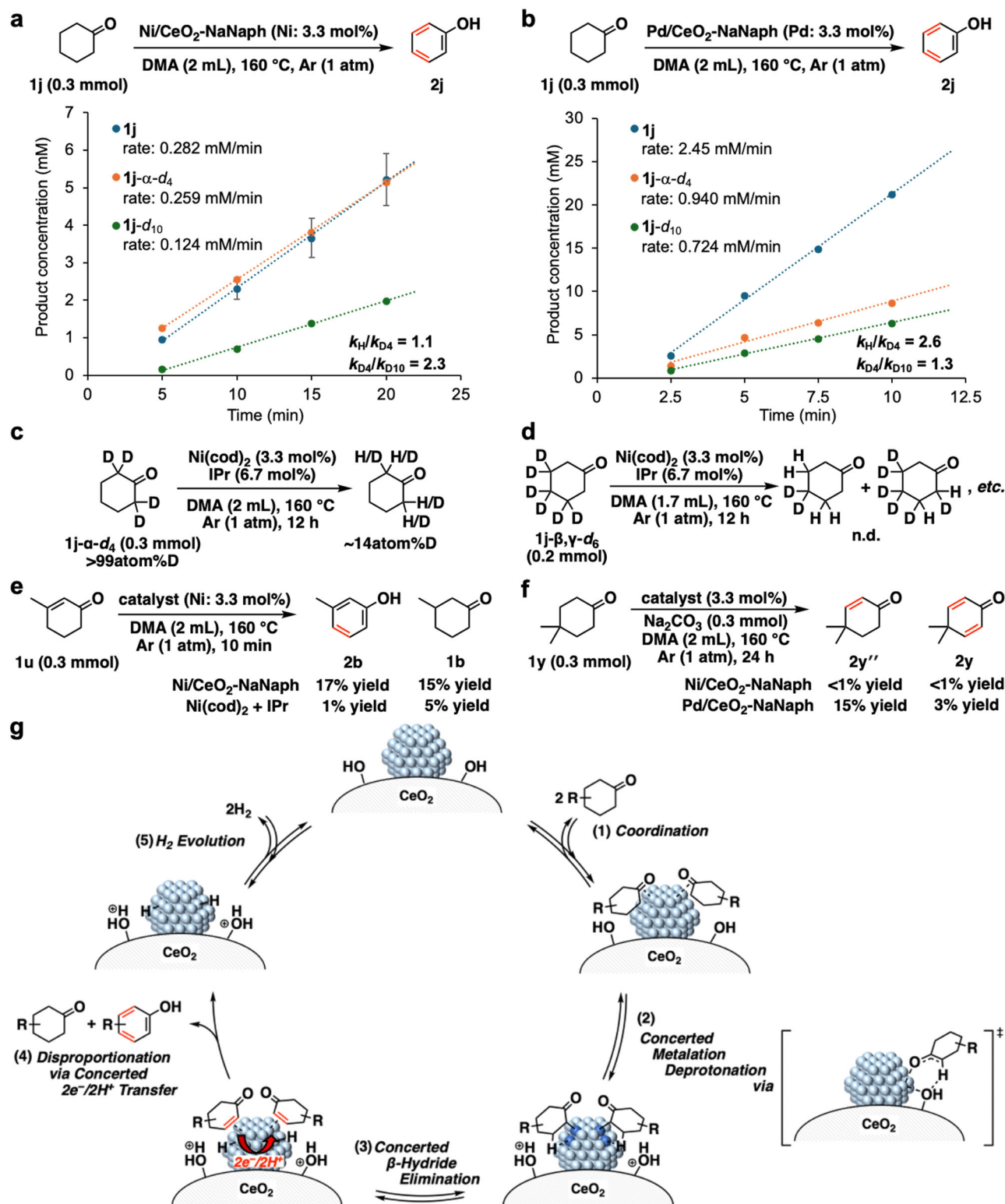


Fig. 5 | Kinetic analyses, control experiments and detailed proposed catalytic cycle. Kinetic isotope effect on acceptorless dehydrogenative aromatization of **1j** in the presence of (a) Ni/CeO₂-NaNaph or (b) Pd/CeO₂-NaNaph. c H/D exchange of **1j**-α-d₄ in the presence of the Ni-IPr complex. d H/D exchange of **1j**-β,γ-d₆

in the presence of the Ni-IPr complex. e Ni/CeO₂-NaNaph- or Ni-IPr-catalysed disproportionation of **1u**. f Acceptorless dehydrogenation of **1y** in the presence of Ni/CeO₂-NaNaph or Pd/CeO₂-NaNaph. g Detailed proposed catalytic cycle of the acceptorless dehydrogenative aromatization of cyclohexanones. Note: n.d. not detected.

unique active site structures of supported nanoparticle catalysts compared to those of mononuclear complexes effectively facilitate organic reactions. We expect that our findings will contribute to the development of novel molecular transformations using supported nanoparticle catalysts in the future.

Methods

Preparation of Ni/CeO₂-NaNaph

First, Ni(OH)_x/CeO₂ was prepared as the precursor according to the following method. The aqueous solution (60 mL) of NiCl₂·6H₂O (118.8 mg, 8.3 mM) with CeO₂ (2.0 g), which was calcinated in the air at

550 °C for 3 h, was vigorously stirred at room temperature. After 15 min, the pH of the solution was adjusted to 11.0 by addition of an aqueous solution of NaOH (1.0 M), and the resulting slurry was stirred overnight. The solid was then filtered off, washed with a large amount of water (3 L) and dried to afford 2.0 g of Ni(OH)_x/CeO₂ as a pale yellow powder. Ni(OH)_x/CeO₂ was stored in an Ar-filled glove-box after being heated at 150 °C for 15 min *in vacuo*.

Then, Ni/CeO₂-NaNaph was prepared as follows. Ni(OH)_x/CeO₂ (1.0 g) was added to the THF solution (12 mL, 0.25 M) of NaNaph and stirred vigorously at room temperature for 90 min. The solid was then filtered off, washed with methanol (20 mL) and diethyl ether (20 mL) and dried under an Ar atmosphere to afford 1.0 g of Ni/CeO₂-NaNaph as black powder (Ni content: 1.38 wt%). Various supported nickel catalysts, such as Ni/Al₂O₃-NaNaph (Ni content: 1.25 wt%), Ni/TiO₂-NaNaph (Ni content: 0.85 wt%), Ni/ZrO₂-NaNaph (Ni content: 1.20 wt%) and Ni/HAP-Nanaph (Ni content: 1.39 wt%) were prepared in the same method.

Catalytic reaction

In an Ar-filled glovebox, Ni/CeO₂-NaNaph (3.3 mol%, 40 mg), cyclohexanones (0.3 mmol), *n*-hexadecane (0.1 mmol, internal standard), DMA (2.0 mL), and a Teflon-coated magnetic stir bar were placed into a pressure tube (volume: ~40 mL). The mixture was stirred at 180 °C for 24 h, then the mixture was cooled down to room temperature. Conversions and product yields were determined by GC analysis using *n*-hexadecane as an internal standard. As for isolation of the products, *n*-hexadecane was not added. After the reaction, the catalyst was removed by simple filtration and the filtrate was concentrated by evaporation, and then subjected to silica-gel column chromatography, giving the pure product. The products were identified by GC-MS, and NMR (¹H and ¹³C).

Data availability

The data generated in this study are provided within the article and its Supplementary Information file. Correspondence and requests for materials should be addressed to K.Y. and T.Yatabe.

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Author contributions

T.Yat. and K.Y. conceived and supervised the project. T.M. performed most of the experiments. T.Yab. performed the XAFS measurements and analysis. All authors contributed to data analysis and discussed the results. T.M. and T.Yat. wrote the manuscript with feedback from K.Y. and T.Yab.

Competing interests

The authors declare no competing interests.

Additional information

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